

When a new peer is connected, it should send the `Discover` message to all the nodes that it is connected to, and each node that receives that message, should response with a `PeerSync` message, the message contains the a list of all the connected peers (inbound + outbound). finally the new message receives the list of the peers and connects to them.

Discover Message Object

```
class Action(Enum):
    ...
    discover = "DISCOVER"
    ...

class DiscoverMessage(BaseModel):
    sender: NodeMetadata
    toAll: bool = False
```

Rules:

- In the `Discover` message object there is a parameter called `toAll: bool` that tells the new node if the discover message should be sent to all the nodes.

Discover Message Response Object

```
class PeerSync(BaseModel):
    sender: NodeMetadata
    connectedPeers: list[NodeMetadata]
    to: NodeMetadata
    date: timestamp
```

Processing The PeerSync Message

When a node receives a `peerSync` message, it should connect to all the nodes listed in the message.

```
def node_message(self, node, data) -> None:
    ...
    if data.get("action") == Action.peerSync.value:
        self.interaction.handlePeerSync(data.get("data"))
    ...
```

```
def handlePeerSync(self, peerSyncMessage: PeerSync) -> None:
    peers = peerSyncMessage.connectedPeers

    for peer in peers:
        host, port = peer.host, peer.port

        self.network.connect(host, port)
```